

I am an environmental economist working on the integrity and design of carbon markets. On the supply side, I price what a carbon credit actually delivers net of leakage, storage duration, and reversal risk; on the demand side, I measure public support for climate and forest policy across Europe. The work speaks directly to the EU Carbon Removals Certification Framework and the design of climate-policy instruments.

Fields Carbon-market design and integrity · climate and environmental economics · non-market valuation · EU climate policy.

EDUCATION

- 2023–present **Ph.D. in Economics**
AgroParisTech, BETA · Nancy & Palaiseau, France
Thesis: *Pricing Credibility: Supply-Side Integrity and Public Acceptability in Nature Markets*.
Supervisors: Jens Abildtrup, Antonello Lobianco. Defence expected December 2026.
- 2020–2022 **M.Sc. in European Forestry**
Erasmus Mundus Double Degree · University of Eastern Finland; University of Lleida; INRAE, BETA · Finland, Spain & France
Double degree: first year in resource economics and policy; second year in spatial and ecological modelling. Master's thesis (100%, with honours) published in *Ecological Economics* (2023). GPA 97%.
- 2016–2020 **B.Sc. in Forest Sciences**
Dresden University of Technology · Dresden, Germany
Forest science with substantial coursework in economics, political science, and philosophy. Bachelor's thesis on small-scale temporal and spatial variation of soil organic carbon stocks (99%). GPA 88%.

POSITIONS

- 2025–present **Visiting Scholar**
Department of Land Economy (C-EENRG), University of Cambridge · Cambridge, UK
Carbon-credit integrity and pricing, with Andreas Kontoleon.
- 2025–present **Research Fellow**
Climate Economics Chair · Paris, France
Carbon-market design and forest-based climate policy, with Philippe Delacote.
- 2023–present **Ph.D. Candidate in Economics**
AgroParisTech, BETA · Nancy & Palaiseau, France
Doctoral research on nature-market credibility: supply-side integrity of carbon credits (leakage, permanence, reversal) and demand-side public acceptability of forest and climate policy across Europe.
- 2021–2024 **Research Associate, Assistant & Intern**
INRAE, BETA; AgroParisTech; BOKU Vienna · France & Austria
Design and fieldwork for discrete choice experiments on forest ecosystem services; France–Austria research collaboration.

2018–2019 **Research Assistant & Intern**

Dresden University of Technology; Spanish National Research Council (CSIC) · Germany & Spain

Lab and field research on soil erosion, bio-based flame retardants, and experimental technosols.

PUBLICATIONS

- 2026 Chiti, T., Rey, A., Abildtrup, J., Böttcher, H., Diaci, J., **Frings, O.**, Lehtonen, A., Pülzl, H., Schindlbacher, A., & Zavala, M. A. A Review of Forest Management Practices Potentially Suitable for Carbon Farming in European Forests. *Journal of Environmental Management*, **398**, 128391.
- 2024 Chiti, T., Rey, A., Abildtrup, J., Böttcher, H., Diaci, J., **Frings, O.**, Lehtonen, A., & Zavala, M. A. Carbon Farming in the European Forestry Sector. *From Science to Policy* 17. European Forest Institute.
- 2023 **Frings, O.**, Abildtrup, J., Montagné-Huck, C., Gorel, S., & Stenger, A. Do Individual PES Buyers Care About Additionality and Free-Riding? A Choice Experiment. *Ecological Economics*, **212**, 107944.
- Forthcoming Chiti, T., Rey, A., Zavala, M. A., Manrique, S., & **Frings, O.** Productive and Biodiverse: Managing Europe's Forests for Economic and Ecological Outcomes. IUFRO Policy Report.
- Working paper **Frings, O.**, Abildtrup, J., Delacote, P., Kontoleon, A., Böttcher, H., Chiti, T., Rey, A., Diaci, J., Lehtonen, A., Pülzl, H., Schindlbacher, A., & Zavala, M. A. *The Net Climate Value of Carbon Credits: Pricing Leakage, Time, and Reversal*.
- Working paper **Frings, O.**, Abildtrup, J., Shanafelt, D. W., Lobianco, A., & Vacik, H. *One Divide, Two Countries: Climate Concern Splits Support for Forest Conservation Programmes*.
- Working paper **Frings, O.** *A Clustering-Initialised Specification Workflow for Latent Class Choice Models*. Companion R package `k1ue` published (R-universe).
- Working paper **Frings, O.** *Public Preferences over Tropical Carbon Offsets: Moral Distance versus Market Distrust in the EU Land-Sector Gap*.
- Working paper Lobianco, A., Besic, N., Bontemps, J.-D., **Frings, O.**, & Hertzog, L. *A Hybrid Machine Learning Approach for the Prediction of Forest Stand Growth Under Different Climate Scenarios*.

POLICY ENGAGEMENT

Invited Expert, ThinkForest

EU Representation of Austria, Brussels · briefed European Commission officials on the EU Carbon Removals Certification Framework Oct. 2024

Invited Roundtable Speaker

Lauterpacht Centre for International Law, Cambridge · climate and biodiversity policy May 2025

Expert Contributor

Co-design the Future(s) of PES in the EU, Online · EU PES scheme design May 2025

Invited Panellist, ThinkForest

Ask a Scientist, Online · carbon farming in the European forest sector April 2025

AWARDS

- 2024 **Highly Commended Paper Award**
8th Workshop on Non-Market Valuation, University of Leeds · UK
- 2023 **Extraordinary Prize for Master's Degrees**
University of Lleida, Spain · best academic record, 2021/22 cohort

GRANTS

- 2025 **FOCUSED Seed Project Grant**
AAP ARTEMIS · €15,000 · €2,000 for thesis survey expansion
- 2024–2026 **Research-Stay Funding, University of Cambridge**
Four competitive grants · €15,300 total
FairCarboN (French National Research Agency, €7,000); DrEAM (University of Lorraine, €5,300); Agreenium EIR-A-DESSE (€2,000); ABIES (AgroParisTech, €1,000).
- 2022 **ORION Master's Grant**
French National Research Agency · €7,000 · scholarship for excellent foreign students
- 2021 **ETSEA Grant**
University of Lleida, Spain · €2,500 · Erasmus Mundus European Forestry scholarship

COMMUNICATIONS

CONFERENCES

XXVII Annual BIOECON Conference
St John's College, Cambridge, UK · oral Sept. 2026

7th World Congress of Environmental & Resource Economists
Cascais, Portugal · oral June 2026

3rd European Carbon Farming Summit
Padua, Italy · oral March 2026

Int. Symposium on Climate, Finance and Sustainability
Paris, France · oral June 2025

LabEx ARBRE Day
INRAE Champenoux, Nancy · oral Oct. 2024

XXV Annual BIOECON Conference
King's College, Cambridge, UK · oral Sept. 2024

26th IUFRO World Congress
Stockholm, Sweden · poster July 2024

8th Workshop on Non-Market Valuation
University of Leeds, UK · oral June 2024

2nd Action versus Inaction Facing Climate Change Conf.
Strasbourg, France · poster June 2024

10th FAERE Annual Conference
Montpellier, France · oral Sept. 2023

28th EAERE Annual Conference
Limassol, Cyprus · oral June 2023

SEMINARS

C-EENRG Research Seminar
University of Cambridge, UK June 2025

TEACHING

Market-Based Instruments & Payments for Ecosystem Services
AgroParisTech, M.Sc. FEN, Nancy · lecturer Nov. 2024

Introduction to Stated-Preference Valuation Methods
AgroParisTech, M.Sc. FEN, Nancy · lecturer Nov. 2024

ADVANCED TRAINING

Over 200 hours of doctoral coursework, including:

Audited lectures in environmental, regional & spatial economics

University of Cambridge, UK · 2025

12th & 13th Agreenium International Research Schools (EIR-A)

INRAE, France · 2024–2025

Summer School for Systems Modelling

IIASA, Laxenburg, Austria · 80 h · July 2024

Advanced Discrete Choice Modelling & Survey Design

Choice Modelling Academy, France · 70 h · 2023

RESEARCH SOFTWARE

klue R package: clustering-based initialisation, multi-start estimation, and segment selection for latent class choice models. github.com/o-frings/klue

net-climate-value Reproducible R pipeline behind *The Net Climate Value of Carbon Credits*: data, calibration, and simulation code. github.com/o-frings/net-climate-value

SKILLS & LANGUAGES

Microeconometrics Discrete choice and random utility models; welfare and willingness-to-pay estimation; maximum-likelihood estimation; experimental and survey design.

Risk & insurance economics Actuarial and survival models for risk pricing; discounting and net present value; Monte Carlo simulation, bootstrap, and sensitivity analysis.

Environmental & resource economics Non-market valuation; payments for ecosystem services; partial-equilibrium market modelling; carbon-market design and accounting.

Machine learning Random forest and gradient boosting; clustering (k-means, Gaussian mixture, hierarchical, PAM); multiple imputation.

Software R (Apollo, mclust, randomForest, xgboost, MICE), Ngene, Julia, $\text{\LaTeX}$, LimeSurvey.

Languages German (native), English (proficient), French (basic).

LEADERSHIP

2019 **Elected Campaign Organiser**

Political youth organisation, Dresden · Saxony state-election campaign; social media for European & city-council campaigns

REFERENCES

- Ph.D. supervisor **Jens Abildtrup** INRAE, BETA, Nancy, France
jens.abildtrup@inrae.fr
- Co-supervisor **Antonello Lobianco** AgroParisTech, BETA, Nancy, France
antonello.lobianco@agroparistech.fr
- Collaborator **Andreas Kontoleon** Department of Land Economy, University of Cambridge, UK
ak219@cam.ac.uk
- Collaborator **Philippe Delacote** INRAE, BETA & Climate Economics Chair, Paris, France
philippe.delacote@inrae.fr